as needed. It only remains to show that $\delta_z = 0$ for all $z \in [0,1]$. By definition of δ_z,

$$\begin{aligned}\delta_z &= \left| \Pr_{(c,r)\sim\hat{p}}[\hat{p}(r \mid c) > z] - \Pr_{(c,r)\sim p}[\hat{p}(r \mid c) > z] \right| \\ &= \left| \sum_c \mu(c) \sum_{r:\hat{p}(r|c)>z} (\hat{p}(r \mid c) - p(r \mid c)) \right|\end{aligned}$$

By definition $\hat{p}(r \mid c) = p(r \mid c)$ everywhere except for $c \in \mathcal{U}, r \in \mathcal{R}_c$. But for each $c \in \mathcal{U}$, $\hat{p}(c,r)$ is constant over $r \in \mathcal{R}_c$, so $\hat{p}(c,r) > z$ for either all $r \in \mathcal{R}_c$ or none of them. Hence the inner sum above is 0 in any case because $\sum_{r\in\mathcal{R}_c} \hat{p}(r \mid c) - p(r \mid c) = 0$ and $\hat{p}(\text{IDK} \mid c) = p(\text{IDK} \mid c)$. □

C Poor-model analysis

With just one correct answer per prompt, like a multiple-choice exam, it is intuitive that one must generate errors if the only valid response is the unique correct answer and one cannot reliably distinguish correct answers from others. For such a simple case, we show the existence of a threshold t with a better bound. In particular, let

$$\text{err}_{\text{iiv}}(\hat{f}_t) := \Pr_{x\sim D}\left[\hat{f}_t(x) \neq f(x)\right], \text{ where } \hat{f}_t(c,r) := \begin{cases} + & \text{if } \hat{p}(r \mid c) > t, \\ - & \text{if } \hat{p}(r \mid c) \leq t. \end{cases}$$

Hence $\hat{f} = \hat{f}_t$ for $t = 1/\min|\mathcal{E}_c|$ and $\hat{f}$ defined in the paper body. We now state and prove a stronger theorem than Theorem 3. Theorem 3 follows immediately from the definition of $\text{opt}(\mathcal{G})$ and the following theorem.

Theorem 4. *Suppose $|\mathcal{V}_c| = 1$ for all $c \in \mathcal{C}$ and let $C = \min_c |\mathcal{E}_c| + 1$ be the number of choices. Then, for all $p, \hat{p}$, there is some threshold $t \in [0,1]$ such that:*

$$\text{err} \geq 2\left(1 - \frac{1}{C}\right)\text{err}_{\text{iiv}}(\hat{f}_t).$$

Note that the proof of Corollary 2 follows immediately from Theorem 4

Proof of Corollary 2. The proof follows immediately from Theorem 4 and the fact that $\text{err}_{\text{iiv}}(\hat{f}_t) = 1/2$ because a classifier $\hat{f}_t$ based on a trigram model cannot distinguish between c_1, c_2. □

We now prove Theorem 4.

Proof of Theorem 4. Consider picking a uniformly random $t \in [0,1]$. We show that:

$$\text{err} \geq 2\left(1 - \frac{1}{C}\right)\mathop{\mathbb{E}}_{t\in[0,1]}[\text{err}_{\text{iiv}}(\hat{f}_t)], \tag{11}$$

This implies that there must exist some threshold $t \in [0,1]$ for which it holds. Note that for uniformly random $t \in [0,1]$,

$$\Pr_{t\in[0,1]}\left[\hat{f}_t(c,r) = +\right] = \hat{p}(r \mid c).$$

First, the expected false positive rate (misclassifications where $\hat{p}(r \mid c) > t$) is:

$$\begin{aligned}
\Pr_{t\in[0,1],x\sim D}\left[\hat{f}_t(x) = +, f(x) = -\right] &= \frac{1}{2}\sum_c \mu(c) \sum_{r\in\mathcal{E}_c} \frac{1}{|\mathcal{E}_c|} \Pr_t\left[\hat{f}_t(c,r) = +\right] \\
&\leq \frac{1}{2}\sum_c \mu(c) \sum_{r\notin\mathcal{A}_c} \frac{1}{C-1}\hat{p}(r \mid c) \\
&= \frac{1}{2(C-1)}\,\mathrm{err}\,.
\end{aligned}$$

Second, let $\mathcal{A}_c = \{a_c\}$ for each c. Then the expected false negative rate is,

$$\begin{aligned}
\Pr_{t\in[0,1],x\sim D}\left[\hat{f}_t(x) = -, f(x) = +\right] &= \frac{1}{2}\sum_c \mu(c) \Pr_t\left[\hat{f}_t(c,a_c) = -\right] \\
&= \frac{1}{2}\sum_c \mu(c)\left(1 - \hat{p}(a_c \mid c)\right) \\
&= \frac{1}{2}\,\mathrm{err}\,.
\end{aligned}$$

Hence the expected misclassification rate, the sum of the expected false positive and negative rates, satisfies:

$$\mathop{\mathbb{E}}_t[\mathrm{err}_{\mathrm{iiv}}(\hat{f}_t)] \leq \frac{1}{2}\left(\frac{1}{C-1} + 1\right)\mathrm{err},$$

which is equivalent to Eq. (11) after rearranging terms. □

D Computationally intractable hallucinations

In this section we provide a stylized example of computational intractability Section 3.4. More natural examples of empirically hard problems that induce hallucinations are examined by Fan et al. (2024) and Tang et al. (2025).

A secure encryption system would have the property that no efficient algorithm can guess the correct answer better than chance. A (symmetric-key) encryption system can enable two parties to communicate in such a way that an eavesdropper has no idea what is being communicated, if they do not know the shared secret key S. Formally, such a setting has sets of messages $\mathcal{M}$, ciphertexts $\mathcal{H}$, an encryption function $e_S : \mathcal{M} \to \mathcal{H}$, and decryption function $d_S : \mathcal{H} \to \mathcal{M}$, such that $d_S(e_S(m)) = m$ for all $m \in \mathcal{M}$.

In the context of hallucinations, let p output (c, r) where $r \in \mathcal{M}$ is uniformly random and the prompt c takes the form "What is the decryption of h?" where $h = e_S(r)$. Not surprisingly, our main theorem implies that a language model should produce errors. In a secure system, without knowing S one cannot distinguish a pair $(m, e_S(m))$ from (m, h) where $m \in \mathcal{M}$ is a uniformly random message and $h \in \mathcal{H}$ is an incorrect (or a uniformly random) ciphertext. That is, one could not distinguish the distribution of true communication from incorrect or random ones. This formulation matches our distribution D which has, with probability 1/2, $x = (e(m), m)$, and with probability 1/2, $x = (h \neq e(m), m)$ where $h \in \mathcal{H} \setminus \{e(m)\}$ is uniformly random. This corresponds to random prompts for μ, and the target function $f(h, r) = +$ iff $h = e(r)$. One form of a standard hardness security definition would be the following (see, e.g., Goldreich, 2001):

Definition 3 (Secure encryption)**.** *Let $\beta \in [0,1]$. Classifier $\hat{f} : \mathcal{X} \to \{+,-\}$ β-breaks the encryption scheme if*

$$\Pr_{x \sim D}[\hat{f}(x) \neq f(x)] \leq \frac{1-\beta}{2}.$$

As mentioned, a random distribution $\hat{p}$ has $\delta = 0$, regardless of t, hence it is easy to have weakly calibrated responses. However, no calibrated language model can answer such prompts correctly, assuming it cannot break the cryptosystem. With these definitions, Theorem 1 immediately implies the following using $|\mathcal{V}_c| = 2$ and $|\mathcal{E}_c| = |\mathcal{M}| - 1$:

Observation 2. *For any $\beta \in [0,1]$ and any language model $\hat{p}$, if the classifier $\hat{f}$ does not β-break the encryption security, then $\hat{p}$ will output erroneous decryptions r with probability at least,*

$$1 - \beta - \frac{2}{|\mathcal{M}| - 1} - \delta.$$

This stylized example illustrates how our reduction applies to computationally hard problems, and how computational hardness from supervised learning parallels computational hardness as a factor for hallucinations.

E Post-training analysis

Below is the short proof of Observation 1.

Proof of Observation 1. It was assumed that $g_c(r) = 0$ for all $r \in \mathcal{A}_c$ and every binary grader g_c is assumed to take on $g_c(r) = 1$ at some value $r \in \mathcal{R}_c \setminus \mathcal{A}_c$. Moreover, since $\mathcal{X}$ was assumed to be finite, there must be some such r which has $\Pr_{g_c \sim \rho_c}[g_c(r) = 1] > 0$. This follows from the union bound:

$$\sum_{r \in \mathcal{R}_c} \Pr_{g_c \sim \rho_c}[g_c(r) = 1] \geq \Pr_{g_c \sim \rho_c}[\exists r \; g_c(r) = 1] = 1.$$

Thus, all $r \in \mathcal{A}_c$ are strictly suboptimal in terms of expected score. □

F Current grading of uncertain responses

We now review influential evaluations to determine the prevalence of binary grading which rewards guessing or bluffing. Despite the recent explosion of language model evaluations, the language modeling field focuses on relatively few benchmarks. Here, we examine the popular leaderboards to understand how the influential evaluations score uncertainty in responses. Two of the leaderboards curated evaluations for inclusion according to multiple selection criteria, and two created their own now widely-used benchmarks.

Table 2 (page 14) shows the ten evaluations selected here. Only one evaluation included in one of the leaderboards, WildBench (Lin et al., 2025), offers minimal credit given for indicating uncertainty. Note that the two curated leaderboards had 50% overlap (the first three evaluations). As further evidence of the attention given to these evaluations, note that Google's latest language model card (Gemini 2.5 Pro, Google DeepMind, 2025) included results for GPQA, MMLU, SWE-bench, HLE, and AIME (similar to MATH L5). OpenAI has similarly published results for GPQA (OpenAI, 2024), MMLU and SWE-bench verified (OpenAI, 2025d), IFEval (OpenAI, 2025c), MATH (OpenAI,